

Security Advisory: Authentication Bypass due to Improper Credential Validation and Recoverable Embedded Credentials in Trueview Security Camera

Overview

The Trueview T18161-AF is a network-enabled security camera used for surveillance applications. As part of a security assessment, hardware reconnaissance was performed to identify accessible interfaces and evaluate the overall security posture of the device.

During the assessment, multiple hardware interfaces were identified that could facilitate access to the device firmware. Firmware acquisition and subsequent analysis were performed to review the device's internal components, configuration artifacts, and authentication mechanisms.

Analysis of the extracted firmware revealed authentication-related artifacts containing embedded credentials. Further review showed that the credentials were stored in a recoverable form within the firmware image. Further evaluation of the authentication mechanism revealed improper credential validation. Authentication attempts using username and password combinations that were not associated with legitimate credentials were accepted by the device, resulting in an authentication bypass condition.

The recovered credentials were subsequently validated against the target device and were accepted by the authentication mechanism, providing access to authenticated functionality.

Deeper analysis of the device's authentication process revealed that credential validation was not being enforced correctly. The device accepted user-supplied username and password combinations without requiring valid authentication credentials.

This weakness may allow unauthorized access to affected devices if the firmware image is obtained and analyzed, thereby undermining the intended authentication controls of the device.

This disclosure follows responsible vulnerability disclosure practices and has been assigned CVE-2026-37270.

Assigned CVE: **CVE-2026-37270**



Vulnerability Summary

The assessment identified multiple weaknesses affecting the device's authentication mechanism.

Analysis of the extracted firmware revealed embedded authentication credentials stored in a recoverable form within firmware artifacts. The credentials could be identified through firmware analysis and were subsequently validated against the target device.

Further evaluation of the authentication process revealed improper credential validation. The device accepted authentication attempts using username and password combinations that were not associated with legitimate credentials, resulting in an authentication bypass condition.

These weaknesses undermine the intended authentication controls of the device and may allow unauthorized access to authenticated functionality.

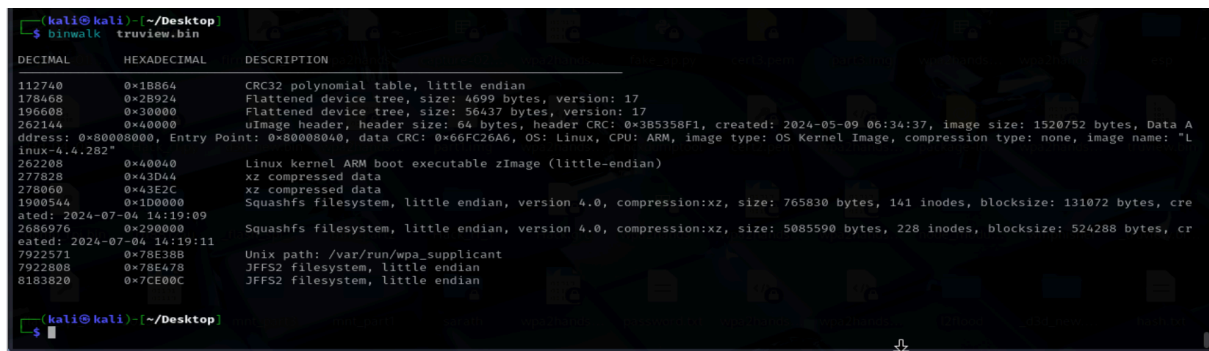
Affected Product

Product: Trueview Security Camera

Model: T18161-AF

Firmware Version: 4.9.60.0

Assessment Methodology



The assessment focused on firmware security and credential management practices.

Firmware contents were reviewed to identify authentication-related artifacts and evaluate how credentials were stored and protected within the device firmware.

The objective of the assessment was to determine whether authentication secrets could be recovered and whether recovered credentials could impact device security.

Findings

Finding 1 – Recoverable Embedded Credentials

```
(kali@kali)-[~/Desktop/truview/_truview.bin.extracted]
$ squashfs-root-0
squashfs-root-0/sbin/rtl8188.sh:10:# Author:
squashfs-root-0/sbin/kill_ipc.sh:4:# Description: update zImage6rootfs under dir1/dir2/ ...
squashfs-root-0/sbin/kill_ipc.sh:5:# Author: xc
squashfs-root-0/sbin/nk_update.sh:4:# Description: update zImage6rootfs under dir1/dir2/ ...
squashfs-root-0/sbin/nk_update.sh:5:# Author: xc
squashfs-root-0/sbin/nk_update.sh:39:mkdir /tmp/root
squashfs-root-0/sbin/nk_update.sh:40:DIR_ROOT_TMP=/tmp/root
squashfs-root-0/sbin/nk_update.sh:41:dd if=/dev/root of=/dev/rootfstmp
squashfs-root-0/sbin/nk_update.sh:43:echo "mkdir -p $DIR_ROOT_TMP"
squashfs-root-0/sbin/nk_update.sh:44:mount -t squashfs /dev/rootfstmp $DIR_ROOT_TMP
squashfs-root-0/sbin/nk_update.sh:47:export LD_LIBRARY_PATH="$DIR_ROOT_TMP/lib"
squashfs-root-0/sbin/nk_update.sh:48:export PATH="$DIR_ROOT_TMP/bin:$DIR_ROOT_TMP/sbin"
squashfs-root-0/sbin/main.sh:25:#start ftp server, dir=root r/w, -t 600s(timeout)
squashfs-root-0/sbin/anyka_ipc.sh:10:# Author: gao_wangsheng
grep: squashfs-root-0/lib/libnsl-3.so.200: binary file matches
grep: squashfs-root-0/lib/libwolfssl.so: binary file matches
grep: squashfs-root-0/lib/libiconv.so.2: binary file matches
grep: squashfs-root-0/lib/libiconv.so.2.5.1: binary file matches
grep: squashfs-root-0/lib/libnsl-genl-3.so.200: binary file matches
grep: squashfs-root-0/lib/libiconv.so: binary file matches
grep: squashfs-root-0/lib/libwolfssl.so.18: binary file matches
grep: squashfs-root-0/bin/hostapd: binary file matches
grep: squashfs-root-0/bin/iwlist: binary file matches
grep: squashfs-root-0/bin/hostapd_cli: binary file matches
grep: squashfs-root-0/bin/aic8800d80/fmacfw_8800d80_u02.bin: binary file matches
grep: squashfs-root-0/bin/wpa_supplicant: binary file matches
grep: squashfs-root-0/bin/anyka_ipc: binary file matches
grep: squashfs-root-0/bin/redirect: binary file matches
squashfs-root-0/bin/onvif_config.xml:8: <need_auth>1</need_auth>
squashfs-root-0/bin/onvif_config.xml:19: <username>[REDACTED]</username>
squashfs-root-0/bin/onvif_config.xml:20: <password>[REDACTED]</password>
squashfs-root-0/bin/onvif_config.xml:21: <userlevel>[REDACTED]istrator</userlevel>
squashfs-root-0/bin/onvif_config.xml:25: <password>[REDACTED]</password>
grep: squashfs-root-0/bin/nk_upgarde: binary file matches
grep: squashfs-root-0/bin/mmcblk: binary file matches
grep: squashfs-root-0/bin/IOTDaemon: binary file matches
grep: squashfs-root-0/bin/wpa_cli: binary file matches
grep: squashfs-root-0/bin/mkdosfs: binary file matches
grep: squashfs-root-0/bin/rtwpriv: binary file matches
grep: squashfs-root-0/bin/iwpriv: binary file matches
```

Analysis of the extracted firmware revealed authentication-related configuration artifacts containing embedded credentials.

The credentials were accessible within the firmware image and could be identified through firmware analysis.

Observations:

- Authentication credentials were present within firmware contents.
- Credentials were recoverable through analysis.
- No effective protection mechanism was observed.
- Firmware disclosure results in credential disclosure.

Finding 2 - Authentication Validation

Recovered credentials were validated against the target device.

The device accepted the recovered credentials and granted access to authenticated functionality.

Observations:

```
kali@kali: ~/~/truview/_truview.bin.extracted/squashfs-root-0/bin
$
Copyright (c) 2003-2025 the FFmpeg developers
built with gcc 15 (Debian 15.2.0-12)
configuration: --prefix=/usr --extra-version=3 --toolchain-hardened --libdir=/usr/lib/x86_64-linux-gnu --incdir=/usr/include/x86_64-linux-gnu --arch=amd64
--enable-gpl --disable-stripping --disable-pocketsphinx --disable-libcaca --disable-libmfx --disable-omx --enable-gnutls --enable-libaom --enable-libass
--enable-libb2b --enable-libcdio --enable-libcodecs2 --enable-libdav1d --enable-libflite --enable-libfontconfig --enable-libfreetype --enable-libfribidi --
enable-libglslang --enable-libgme --enable-libgsm --enable-libharfbuzz --enable-libmp3lame --enable-libmysofa --enable-libopenjpeg --enable-libopenmpt --en
able-libopus --enable-librubberband --enable-libshine --enable-lisnappy --enable-libsoxr --enable-lispeex --enable-libtheora --enable-libtwolame --enable-
libvidstab --enable-libvorbis --enable-libvpx --enable-libwebp --enable-libx265 --enable-libxml2 --enable-libxvid --enable-libzimg --enable-opengl --enabl
e-openc1 --enable-opengl --disable-sndio --enable-libvpl --enable-libdc1394 --enable-libdrm --enable-libiec61883 --enable-chromaprint --enable-frei0r --ena
ble-ladspa --enable-libluray --enable-libvdpau --enable-libvdpau --enable-libjack --enable-libjxl --enable-libpulse --enable-librabbittmq --enable-libri
able-librav1e --enable-librsvg --enable-shared
WARNING: library configuration mismatch
swresample configuration: --prefix=/usr --extra-version=2 --toolchain-hardened --libdir=/usr/lib/x86_64-linux-gnu --incdir=/usr/include/x86_64-linux-gnu
--arch=amd64 --enable-gpl --disable-stripping --disable-pocketsphinx --disable-libmfx --disable-omx --enable-gnutls --enable-libaom --enable-libass --enab
le-libb2b --enable-libcdio --enable-libcodecs2 --enable-libdav1d --enable-libflite --enable-libfontconfig --enable-libfreetype --enable-libfribidi --enable-
libglslang --enable-libgme --enable-libgsm --enable-libharfbuzz --enable-libmp3lame --enable-libmysofa --enable-libopenjpeg --enable-libopenmpt --enable-l
ibopus --enable-librubberband --enable-libshine --enable-lisnappy --enable-libsoxr --enable-lispeex --enable-libtheora --enable-libtwolame --enable-libvi
dstab --enable-libvorbis --enable-libvpx --enable-libwebp --enable-libx265 --enable-libxml2 --enable-libxvid --enable-libzimg --enable-opengl --enable-open
c1 --enable-opengl --disable-sndio --enable-libvpl --enable-libdc1394 --enable-libdrm --enable-libiec61883 --enable-chromaprint --enable-frei0r --enable-la
dspa --enable-libluray --enable-libcaca --enable-libvdpau --enable-libvdpau --enable-libjack --enable-libjxl --enable-libpulse --enable-librabbittmq --e
nable-librist --enable-lisrt --enable-lisssh --enable-lisvtavi --enable-libx264 --enable-libzmq --enable-libzvi --enable-lv2 --enable-sdl2 --enable-libp
lacebo --enable-librav1e --enable-librsvg --enable-shared
libavutil 60. 8.100 / 60. 8.100
libavcodec 62. 11.100 / 62. 11.100
libavformat 62. 3.100 / 62. 3.100
libavdevice 62. 1.100 / 62. 1.100
libavfilter 11. 4.100 / 11. 4.100
libswscale 9. 1.100 / 9. 1.100
libswresample 6. 1.100 / 6. 1.100
Metadata:
title : session
Duration: N/A, start: 0.000000, bitrate: N/A
Stream #0:0: Audio: pcm_alaw, 8000 Hz, mono, s16, 64 kb/s
Stream #0:1: Video: hevc (Main), yuv420p(tv, unknown/bt709/unknown), 2304x1296, 15 fps, 28.58 tbr, 90k tbn
34.53 A-V: -0.350 fd= 7 aq= 273KB vq= 3925KB sq= 0B B sq= 0B
kali@kali: ~/~/truview/_truview.bin.extracted/squashfs-root-0/bin
$
```

- Authentication succeeded using recovered credentials.
- Protected services accepted the credentials.
- Authenticated functionality became accessible.
- Legitimate user credentials were not required.

Finding 3 – Improper Authentication Validation

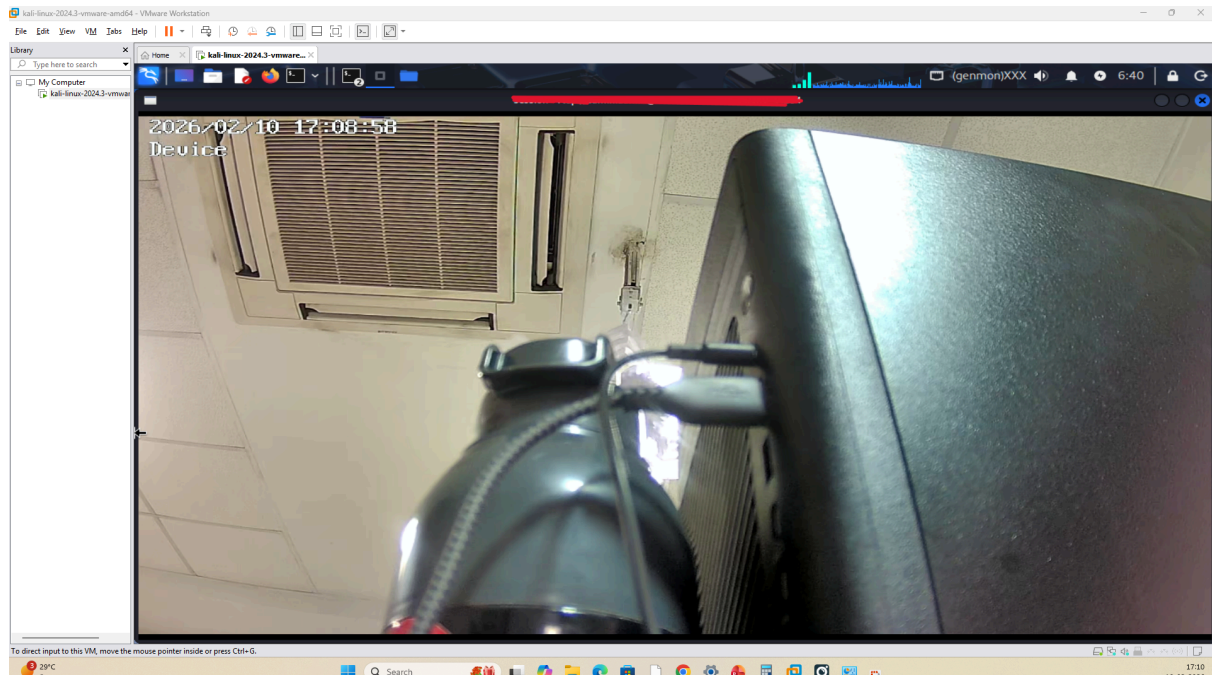
Further testing of the authentication mechanism revealed that credential validation was not performed correctly.

The device accepted authentication attempts using username and password combinations that were not associated with legitimate credentials.

Observations

- Authentication succeeded using non-legitimate credentials.
- Access to authenticated functionality was granted.
- Authentication controls could be bypassed without valid credentials.

Impact



An attacker capable of obtaining the firmware image may recover embedded authentication credentials and use them to authenticate to affected devices. The security impact is further increased by improper authentication validation. Testing demonstrated that the device accepted authentication attempts using non-legitimate credentials, effectively bypassing the intended authentication controls.

Potential impact includes:

- Authentication Bypass
- Unauthorized Access to Device Functionality
- Exposure of Protected Services
- Access Using Non-Legitimate Credentials
- Reduced Trust in Authentication Controls

Recommendations

For Vendors

- Eliminate hard-coded credentials from firmware.
- Generate unique credentials for each device.

- Protect authentication secrets using secure storage mechanisms
- Conduct firmware reviews to identify embedded credentials and other sensitive information.
- Implement proper credential validation mechanisms and ensure authentication requests are verified against legitimate credentials before access is granted.

For Users

- Apply vendor-provided firmware updates when available.
 - Restrict exposure of device management interfaces.
 - Follow vendor security recommendations and best practices.
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Conclusion

The assessment identified recoverable authentication credentials embedded within the firmware of the affected device.

The assessment revealed both recoverable embedded credentials and improper authentication validation within the device. Authentication testing demonstrated that access to authenticated functionality could be obtained using credentials that were not associated with legitimate users, resulting in an authentication bypass condition.

This vulnerability highlights the importance of secure credential management practices and the risks associated with embedding authentication secrets within firmware images.

Disclaimer

This research was conducted solely for educational and defensive security purposes.

All testing was performed in a controlled environment on authorized equipment.

The findings were responsibly disclosed prior to public release.

The information presented in this advisory is intended to promote security awareness and support remediation efforts.

Discoverers

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